$\binom{5}{2} = 10$

1, 2, 3

$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

$\left(\frac{3}{4}\right)^3 = \frac{27}{64}$ $1 - \frac{27}{64} = \frac{37}{64}$ $\frac{37}{64} \cdot \frac{1}{2}$

$1 - \frac{1}{4} = \frac{3}{4}$

10 pairs, $\frac{37}{128} \times 10 = \frac{370}{128} = \frac{185}{64}$